Devansh Shah

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EDUCATION

M.S. in Computer Science, New York University | GPA: 3.78/4.00 Sep 2025 – May 2027
B.Tech. in Computer Science, Nirma University | GPA: 8.51/10.00 Oct 2020 – May 2024

TECHNICAL SKILLS

Languages	Python, C++, SQL
ML / Deep Learning	PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, CNNs, Transformers, fine-tuning
LLM / Retrieval	RAG, embeddings, reranking, LLM evaluation, red teaming, agentic systems, vLLM
Vision / Speech	Face recognition, OCR, vision-language models, speech translation
Systems / Cloud	CUDA, GPU performance, FastAPI, Docker, Kubernetes, AWS, GCP, Weaviate, Git

EXPERIENCE

Data Science Intern May 2026 – Aug 2026
NFP, an Aon Company *New York, NY*

- Built a hybrid classifier combining embeddings, logistic regression, and prompt-tuned Gemma 4 to separate construction-phase claims from post-construction defects; **89% PR-AUC** vs. **57%** baseline on 11,000+ imbalanced claims.
- Engineered a semantic clustering pipeline using embeddings, HDBSCAN, and Gemma 4 to group claims by accident similarity without labeled data or a fixed cluster count; refined noise points and thresholds via code-driven human-in-the-loop feedback, helping underwriters uncover recurring loss patterns across diverse risks and reduce losses.
- Trained a Random Forest on imbalanced monthly loan data to predict one-year delinquency, achieving **85% PR-AUC**; supports proactive identification of at-risk accounts and reserve estimation.

Founding Engineer Jan 2026 – June 2026
LitmusAI *New York, NY*

- Co-founded LitmusAI, a platform for evaluating AI agent behavior and enabling insurance underwriting for AI systems.
- Built a Python SDK with callback-based instrumentation and trace-level observability; integrated drift detection and regression testing into CI/CD across **10 attack/failure scenarios**, surfacing **behavioral drift** in a pilot's agent.
- Led user discovery and onboarding for **5 pilots**, gathered customer feedback to refine LitmusAI's testing workflows.

Machine Learning Engineer Jul 2024 – Jul 2025
Blueinfy Solutions *Ahmedabad, India*

- Built an LLM red-teaming framework by fine-tuning Llama 3.1 and Phi-3.5 Mini on synthetic data from past human-written attacks to generate adversarial prompts, run attacks, and evaluate vulnerabilities across **OWASP Top 10 LLM risks**; used by internal teams and clients, cutting assessments from **8 hrs to ∼4 hrs** at **15%** false positives.
- Deployed a fully automated agentic system using Gemma 3 27B locally to analyze website text and screenshots for sensitive-data exposure across **10,000+ domains**, generating security reports for CISO review.

SELECTED PROJECTS

- **Multilingual Speech Translation** Launch
  Prepared multilingual training data and aligned speech-text pairs; fine-tuned a Gujarati-to-English speech translation model, using human review and an LLM tuned to domain-specific nuances for error analysis and retraining, **cutting critical domain errors** from **24% to <2%**. The translation system serves **1.2M+ users across 190+ countries**.
- **Wisdom Multimodal Search and RAG** App
  Built an end-to-end multilingual hybrid retrieval across webpages, articles, audio, video, and PDFs, combining keyword and semantic search, timestamp-level indexing, and LLM-based reranking; **92%** recall@5 on 250 popular queries with labeled expected results, **sub-1.5s** median latency. Serves **10,000+** monthly users.
- **Autonomous Drone Obstacle Avoidance with Selective VLM Reasoning**
  Built a simulated drone obstacle-avoidance pipeline with a confidence-based gate that escalates uncertain detections to Qwen VLM reasoning, achieving **95% end-to-end accuracy** at **45% lower latency** than always-on VLM inference.
- **LLM-Assisted CUDA Kernel Performance Prediction**
  Built a CUDA kernel performance predictor combining static code features, LLM-derived code semantics, and GPU hardware characteristics; **5% MAPE** on **100 held-out kernels**, **20% lower** than baseline.